Handedness-Aware Cross-Attention SVD for Multimodal Dental Registration

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Abstract. Registering intraoral scans to CBCT is challenging due to modality differences and varying coordinate handedness. We propose a handedness-aware framework for MICCAI STS 2026 Task 2. The method normalizes STL and CBCT surfaces, encodes them via modality-specific PointNet++, and establishes soft correspondences with bidirectional cross-attention. A differentiable Procrustes layer estimates $O(3)$ transformations, enabling both rotations and reflections. A bounded residual branch corrects translation only. We adopt a two-stage semi-supervised strategy. The PointNet++ encoders are first pre-trained on the unlabeled training cases using contrastive self-supervision and are then fine-tuned on the limited labeled cases using ground-truth rigid transformations. Training combines point, rotation, translation, and residual losses; inference uses five-sample test-time aggregation. On a development-only stratified split of six held-out patients (24/6), the upper model achieved 15.402 mm translation and 9.287° rotation error; the lower model achieved 27.612 mm and 12.640°. Jaw-averaged errors were 21.507 mm and 10.963°. These component choices were not independently isolated on this small split. On the official final test set, the method achieved a translation error of 39.3343 mm and a rotation error of 11.6964°.

Keywords: Dental registration · CBCT · Intraoral scan · Cross-attention.

1 Introduction

CBCT provides bone and root structure, whereas IOS provides high-resolution crowns. Their registration supports treatment planning and surgical guidance. STS Task 2 evaluates predicted 4×4 transformations by mean translation and rotation errors. The main difficulties are modality mismatch, thresholded-CBCT

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distractors, initialization-sensitive ICP [1], centre-specific NIfTI conventions, and target transforms with determinant -1, which standard $SO(3)$ Kabsch cannot represent. PointNet++ [3,4], attention, and contrastive learning [5,2] provide point encoding and unlabeled pre-training; prior STS work addresses segmentation rather than multimodal registration [6,7,8].

Our contributions are fourfold:

1. A geometry-preserving STL/CBCT preprocessing pipeline with world-coordinate recovery.
2. A dual-branch PointNet++ and bidirectional cross-attention head for soft, confidence-weighted correspondences.
3. A handedness-aware weighted Procrustes layer that estimates $O(3)$ transforms and selects the determinant from the NIfTI affine signature.
4. Bounded translation refinement, determinant-stratified validation, EMA selection, and test-time aggregation in the submitted system; independent gains are not claimed.

2 Method

2.1 Problem formulation and system overview

Let $S = \{s_i\}_{i=1}^{N_s} \subset \mathbb{R}^3$ be points sampled from an upper- or lower-jaw IOS mesh and let $C = \{c_j\}_{j=1}^{N_c} \subset \mathbb{R}^3$ be points extracted from the corresponding CBCT volume. The objective is to estimate

$$\mathbf{T}_{\text{world}} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\top & 1 \end{bmatrix}, \qquad \mathbf{R}^\top\mathbf{R} = \mathbf{I}, \quad \det(\mathbf{R}) \in \{-1, +1\}, \tag{1}$$

such that transformed IOS points align with CBCT. Upper and lower jaws use separate checkpoints with the same architecture and loss; Fig. 1 summarizes the pipeline.

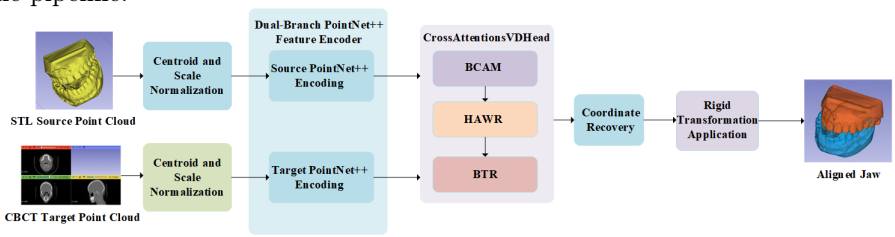


Fig. 1. Complete Task 2 registration framework.

2.2 Geometry-preserving point-cloud preprocessing

IOS and CBCT surfaces. The STL is area-sampled to $N_s = 4096$ points and centred at its AABB centre $\boldsymbol{\mu}_s$. CBCT voxels above 800 HU (with optional one-voxel erosion) are mapped to world coordinates using the NIfTI affine $\mathbf{A}$; the centre and scale are computed before cropping. The lowest 8% of the upper-jaw

z-range is removed, the lower jaw is not cropped, and up to 8192 points are sampled with jaw-specific scale conventions.

Let $a > 0$ denote the selected scale. The network inputs are:

$$\tilde{\mathbf{s}}_i = \frac{\mathbf{s}_i - \boldsymbol{\mu}_s}{a}, \qquad \tilde{\mathbf{c}}_j = \frac{\mathbf{c}_j - \boldsymbol{\mu}_c}{a}. \tag{1}$$

If the network predicts normalized rotation $\mathbf{R}$ and translation $\mathbf{t}_n$, the world-space translation is recovered as:

$$\mathbf{t} = a\mathbf{t}_n + \boldsymbol{\mu}_c - \mathbf{R}\boldsymbol{\mu}_s. \tag{2}$$

2.3 Dual-branch PointNet++ encoding

Separate, unshared PointNet++ encoders process the clouds using three set-abstraction layers (FPS ratios 0.5, 0.25, 1.0; radii 0.2, 0.4, 1.0; widths 32, 64, 128; at most 64 neighbors per centroid).

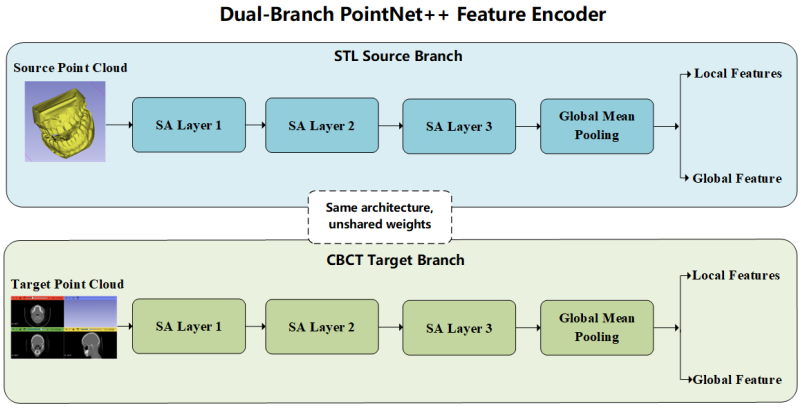


Fig. 2. Dual-branch PointNet++ feature encoder.

For 4096 IOS and 8192 CBCT points, the encoders output 512 and 1024 points, respectively, with 128-dimensional features. Global mean pooling gives $\mathbf{g}_s, \mathbf{g}_c \in \mathbb{R}^{128}$ for translation correction. Encoders are initialized by contrastive pre-training on 300 unlabeled cases; each modality uses two views with rigid perturbations (up to 30° and 0.1 normalized translation), jitter (0.01), 10% dropout, and resampling:

$$\mathcal{L}_{\text{pre}} = \mathcal{L}_{\text{NTX}}(S_1, S_2) + \mathcal{L}_{\text{NTX}}(C_1, C_2) + 0.5\,\mathcal{L}_{\text{NTX}}(S_1, C_1), \tag{3}$$

with temperature 0.1 and 64-dimensional projection heads; only encoder weights are transferred.

2.4 CrossAttentionSVDHead Module

The registration head has three stages: cross-attention matching (CAMM), handedness-aware weighted Procrustes alignment (HAWP), and bounded translation refinement (BTR).

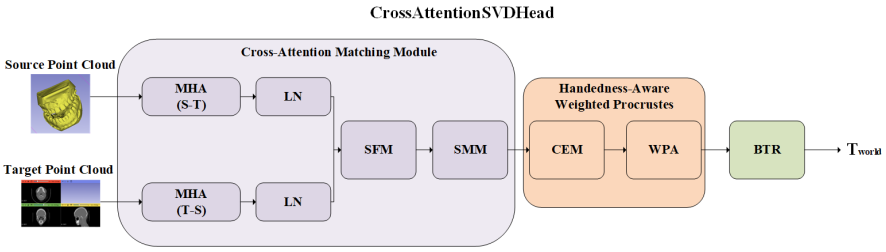


Fig. 3. CrossAttentionSVDHead.

Let $\mathbf{F}_s \in \mathbb{R}^{N_s' \times d}$ and $\mathbf{F}_c \in \mathbb{R}^{N_c' \times d}$ denote the encoded source and target features, with $d = 128$, $N_s' = 512$, and $N_c' = 1024$. The Cross-Attention Matching Module (CAMM) applies four-head cross-attention in both directions:

$$\mathbf{F}_s' = \mathrm{LN}\big(\mathbf{F}_s + \mathrm{MHA}(\mathbf{F}_s, \mathbf{F}_c, \mathbf{F}_c)\big), \tag{4}$$

$$\mathbf{F}_c' = \mathrm{LN}\big(\mathbf{F}_c + \mathrm{MHA}(\mathbf{F}_c, \mathbf{F}_s, \mathbf{F}_s)\big). \tag{5}$$

The Similarity Fusion Module (SFM) projects the attended features into a shared normalized space:

$$Q_s = \ell_2(\mathrm{Linear}_s(F_s')) \tag{6}$$

$$Q_c = \ell_2(\mathrm{Linear}_c(F_c')) \tag{7}$$

$$M = \frac{Q_s Q_c^\top}{\tau}, \quad \tau = 0.07 \tag{8}$$

The Soft Matching Module (SMM) converts $\mathbf{M}$ into a bidirectional, renormalized correspondence distribution:

$$M_{ij} = \frac{Q_s^{(i)} \cdot Q_c^{(j)}}{\tau}. \tag{9}$$

$$P_{ij} = \frac{\mathrm{softmax}_j(M_{ij})\,\mathrm{softmax}_i(M_{ij})}{\sum_k \mathrm{softmax}_k(M_{ik})\,\mathrm{softmax}_i(M_{ik}) + \epsilon}. \tag{10}$$

Here, i, j, k index source points, target points, and the target summation; row and column softmaxes retain mutually agreeing pairs. $\epsilon = 10^{-8}$ prevents division by zero.

The soft corresponding CBCT position for source point i is the probability-weighted expectation over target coordinates:

$$\hat{\mathbf{c}}_i = \sum_j P_{ij} \tilde{\mathbf{c}}_j. \tag{11}$$

The Confidence Estimation Module (CEM) down-weights ambiguous matches using peak probability and entropy, $H_i = -\sum_j P_{ij} \log(P_{ij} + \epsilon)$:

$$w_i = \max_j P_{ij} \cdot \exp(-H_i) = \max_j P_{ij} \cdot \exp\left(\sum_j P_{ij} \log(P_{ij} + \epsilon)\right). \tag{12}$$

The weights are passed to weighted Procrustes alignment. Weighted centroids and cross-covariance are

$$\boldsymbol{m}_s^w = \frac{\sum_i w_i \tilde{\mathbf{s}}_i}{\sum_i w_i}, \qquad \boldsymbol{m}_c^w = \frac{\sum_i w_i \hat{\mathbf{c}}_i}{\sum_i w_i}. \tag{13}$$

$$\mathbf{K} = \sum_i w_i (\tilde{\mathbf{s}}_i - \boldsymbol{m}_s^w)(\hat{\mathbf{c}}_i - \boldsymbol{m}_c^w)^\mathsf{T}. \tag{14}$$

SVD gives $\mathbf{K} = \mathbf{U}\boldsymbol{\Sigma}\mathbf{V}^\mathsf{T}$. The unconstrained solution is $\mathbf{Z}_0 = \mathbf{V}\mathbf{U}^\mathsf{T}$, followed by the handedness correction:

$$\mathbf{R} = \mathbf{V} \operatorname{diag}(1, 1, d_{\text{exp}} \det(\mathbf{Z}_0))\mathbf{U}^\mathsf{T}, \tag{15}$$

where $d_{\text{exp}} \in \{-1, +1\}$ is from ground-truth during training and from the NIfTI affine signature during inference. The base translation is $\mathbf{t}_0 = \boldsymbol{m}_c^w - \mathbf{R}\boldsymbol{m}_s^w$.

The BTR keeps the SVD rotation fixed. A 32-unit MLP receives $\mathbf{g}_s$, $\mathbf{g}_c$, their difference, $\mathbf{t}_0$, the nine entries of $\mathbf{R}$, and the normalized source centre, and predicts a residual clipped to $\|\Delta\mathbf{t}\|_2 \leq 0.20$:

$$\mathbf{t}_n = \mathbf{t}_0 + \Delta\mathbf{t}, \qquad \|\Delta\mathbf{t}\|_2 \leq 0.20. \tag{16}$$

The zero-initialized MLP starts from the SVD solution; detached inputs isolate the residual from encoder and rotation gradients. World translation is recovered as:

$$\mathbf{t} = a\mathbf{t}_n + \boldsymbol{\mu}_c - \mathbf{R}\boldsymbol{\mu}_s. \tag{17}$$

The final transformation matrix is then assembled as

$$\mathbf{T}_{\text{world}} = \begin{bmatrix} \mathbf{R} & \mathbf{t} \\ \mathbf{0}^\mathsf{T} & 1 \end{bmatrix}. \tag{18}$$

This assembled matrix is the head output.

2.5 Supervision Objectives

Use of the partial-label subset. The limited labeled subset is used for supervised fine-tuning after self-supervised pre-training. Upper- and lower-jaw 4×4 matrices supervise separate jaw models; "partial label" denotes the number of labeled cases, not incomplete annotations.

The pose loss combines point alignment, rotation, and translation errors:

$$\mathcal{L}_{\text{pose}} = 1.0\mathcal{L}_{\text{point}} + 2.0\mathcal{L}_{\text{rot}} + 0.02\mathcal{L}_{\text{trans}}. \tag{19}$$

$\mathcal{L}_{\text{point}}$ is point MSE, $\mathcal{L}_{\text{trans}}$ is Euclidean translation error (mm), and $\mathcal{L}_{\text{rot}}$ uses atan2 for same-handed and trace/arccos for opposite-handed transforms.

Neighborhood correspondence supervision has weight 0.1 and encourages matches within 5 mm; the residual Smooth-L1 loss has weight 1.0. The complete objective is:

$$\mathcal{L} = \mathcal{L}_{\text{pose}} + 0.1\mathcal{L}_{\text{corr}} + 1.0\mathcal{L}_{\text{res}}. \tag{20}$$

Optional saliency, direct-pose, SAM, and ICP branches are disabled.

2.6 Inference and Test-Time Aggregation

Five random seeds generate candidates per case. The medoid under a joint translation/rotation metric is selected; candidates within 20 mm and 15° are retained, rotations are matrix-averaged and projected to $O(3)$ by SVD, and translations use the median. Outputs are saved as `upper_gt.npy` and `lower_gt.npy`; ICP, SAM, and saliency are disabled.

3 Experiments

3.1 Dataset and evaluation metrics

Task 2 provides 30 labelled, 300 unlabelled, and 50 public-validation cases with upper/lower 4×4 matrices. Because public labels were unavailable locally, the 30 labelled cases were split once into 24 training and 6 validation patients (seed 2026), stratified by determinant sign, with no overlap.

For predicted and ground-truth transformations $\hat{\mathbf{T}}$ and $\mathbf{T}^*$, translation error is

$$e_t = \|\hat{\mathbf{t}} - \mathbf{t}^*\|_2 \quad \text{(mm)}. \tag{21}$$

For a handedness-aware and numerically valid trace angle, let $d = \det(\hat{\mathbf{R}})\det(\mathbf{R}^*)$ and $\mathbf{D}_d = \text{diag}(1, 1, d)$. We form $\mathbf{Q} = \hat{\mathbf{R}}\mathbf{D}_d\mathbf{R}^{*\mathsf{T}} \in SO(3)$ and report the challenge trace angle,

$$e_r = \frac{180}{\pi} \arccos\left(\text{clip}\left(\frac{\text{tr}(\mathbf{Q}) - 1}{2}, -1, 1\right)\right) \quad \text{(degrees)}. \tag{22}$$

We report patient means for each jaw. Handedness accuracy (matching determinant signs) is diagnostic only and is not an official ranking metric.

3.2 Implementation Details

Environment: Table 1 lists the Ubuntu 22.04, Xeon 8176, 125 GB RAM, RTX 4090 D, CUDA 12.1, and PyTorch 2.5.1 environment.

Table 1. Development and training environment.

System	Ubuntu 22.04.5 LTS
CPU	Intel Xeon Platinum 8176 @ 2.10 GHz
RAM	125 GB
GPU	One NVIDIA GeForce RTX 4090 D (24 GB)
CUDA version	12.1
Programming language	Python 3.9.25
Deep learning framework	PyTorch 2.5.1
Key libraries	NumPy, NiBabel, trimesh, SciPy
Numerical modes	BF16 autocast; TF32 matrix operations

Training: Upper and lower models are optimized separately; lower-jaw training additionally uses the 24 upper-jaw samples (48 samples per epoch), while validation and inference remain jaw-specific.

Source perturbations warm over 15 epochs from normalized translation 0.02/rotation 5° to 0.10/20°; target points are unchanged.

We use AdamW (learning rate 10^{-4}, weight decay 10^{-4}), gradient clipping at 5, cosine annealing to 10^{-6}, and EMA decay 0.995. EMA checkpoints are validated at epoch 1, every 2 epochs, and the final epoch using score $0.1\bar{e}_t + 0.1\bar{e}_r$; inference loads one `best_model.pth` per jaw.

Table 2. Supervised registration training protocol

Pre-trained model	Modality-specific PointNet++ contrastive encoders
Input size	4096 IOS points; 8192 CBCT points
Batch size	4
Total epochs	200 per jaw
Optimizer	AdamW, weight decay 10^{-4}
Initial/minimum learning rate	10^{-4} / 10^{-6}
Learning-rate schedule	Cosine annealing over 200 epochs
Contrastive pre-training time	Upper: 17.24 h; lower: 17.41 h
Supervised training time	Upper: 221.74 s; lower: 379.04 s
Loss	Pose + 0.1× correspondence + residual
Number of model parameters	Approximately 0.26M
Number of FLOPs	Approximately 8.06 GFLOPs/sample
Model selection	Minimum $0.1\bar{e}_t + 0.1\bar{e}_r$ on EMA validation model

4 Results and Discussion

4.1 Internal Validation Results

Table 3 reports the selected models: upper 15.402 mm/9.287° and lower 27.612 mm/12.640°.

Both models achieve 100% handedness accuracy, confirming that determinant selection avoids the forced-$SO(3)$ failure mode. The jaw-averaged errors are 21.507 mm and 10.963°.

Table 3. Internal validation results on the six-patient split.

Jaw	Translation (mm) ↓	Rotation (°) ↓	Handedness (%) ↑
Upper	**15.402**	**9.287**	100
Lower	27.612	12.640	100
Jaw average	21.507	10.963	100

The selected upper checkpoint improves from 16.530 mm (base SVD) to 15.402 mm after residual correction; this comparison is descriptive. The lower jaw remains harder because of smaller coverage and more distractors.

4.2 Development Ablations

Table 4 is descriptive: rows change multiple settings on the same six-patient split and therefore do not identify independent component effects. The compact residual has the lowest upper translation, while the final row has the best upper rotation.

Table 4. Development ablations on the internal validation split.

Configuration	Upper e_t	Upper e_r	Lower e_t	Lower e_r
Handedness-aware SVD	22.214	9.915	28.981	15.204
Decoupled target saliency	22.094	9.822	37.593	**10.708**
Clean SVD objective	22.732	9.932	28.311	15.154
Translation residual	15.501	10.635	29.483	11.392
Compact residual	**14.976**	9.617	28.156	14.283
Paired-jaw training data	15.318	9.513	**26.860**	12.721
Final independent-paired model	15.402	**9.287**	27.612	12.640

e_t is mean translation error in mm and e_r is mean rotation error in degrees. Each row reports the checkpoint minimizing that experiment's combined selection score.

Saliency, ICP, and SAM are disabled; the rows are exploratory rather than causal.

4.3 Qualitative Analysis

Because public-validation ground truth was unavailable, qualitative inspection is illustrative. In cases 076, 176, and 219, IOS surfaces follow the dental arches with consistent orientation and plausible overlap in all three views; complete crowns provide useful cues for cross-attention matching.

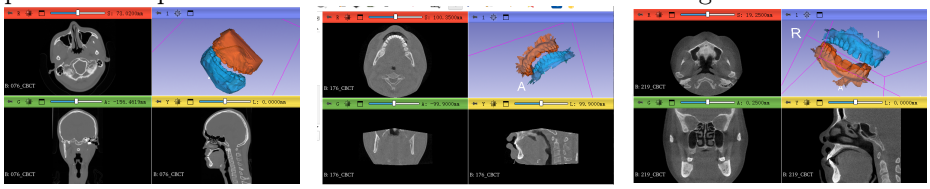


Fig. 4. Successful public-validation cases: 076, 176, and 219.

Cases 057 and 058 show global pose failures, likely caused by retained non-dental structures, incomplete or low-resolution crowns, and repetitive tooth patterns. Independent jaw models cannot correct an error in the other arch. These figures show representative modes, not quantitative generalization evidence.

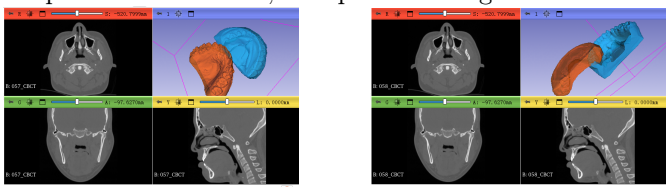


Fig. 5. Failed public-validation cases: 057 and 058.

4.4 Official Validation and Test Results

Batch inference produced matrices for all 50 public-validation cases. Labels were unavailable locally, so official validation and final-test scores came from the challenge server; the six-case split was development-only (Table 5).

Table 5. Official challenge evaluation results returned by the evaluation server.

Split	Translation (mm) ↓	Rotation (°) ↓
Public validation	34.6954	10.9003
Final test	39.3343	11.6964

4.5 Limitations and Future Work

Limitations are the 30 labelled cases, six-patient holdout, non-factorial ablations, centre-specific affine conventions, and fixed-intensity thresholding. Future work includes efficient uncertainty weighting and a shared jaw-conditioned model.

5 Conclusion

The final system combines physical-coordinate preprocessing, PointNet++, cross-attention, soft correspondences, weighted $O(3)$ SVD, bounded translation refinement, and five-sample aggregation. The six-patient split does not support independent causal claims; official scores are in Table 5.

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